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0 1 1 2 3 5 8

Problem

Nth Fibonacci Number (Space Optimised Tabulation)

$n = 6$

	0	1	2
dp	5	8	8

a	b	c
~~0~~	~~1~~	1
~~1~~	~~1~~	2
~~1~~	~~2~~	3
~~2~~	~~3~~	5
3	5	8

5 times

loop → n-1 times

$$\boxed{\begin{aligned} dp[2] &= dp[0] + dp[1] \\ dp[0] &= dp[1] \\ dp[1] &= dp[2] \end{aligned}}$$

$$dp[i] = dp[i-1] + dp[i-2]$$

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Problem

Stickler Thief (Space Optimised Tabulation)

money	6	10	1	2	15
dp	6	10	10	12	25

10	12	25

$dp[2] = max(dp[1], money[i] + dp[0])$

$dp[0] = dp[1]$

$dp[1] = dp[2]$

$dp[i] = max(dp[i-1], money[i] + dp[i-2])$

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Problem

Number of Paths (Space Optimised Tabulation)

m=5
~~m=4~~; n=7

1	1	1	1	1	1	1
1	2	3	4	5	6	7
1	3	6	10	15	21	28
1	4	10	20	35	56	84
1	5	15	35	70	126	210

O(m*n) space

$dp[1][j] = dp[1][j-1] + dp[0][j]$

0	1	4	10	20	35	56	84
1	1	5	15	35	70	126	210

space = 2*n = O(n)

time = 2* prev
as ↓
same^ prev

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Problem

Friends Pairing Problem

$friends(n) = friends(n-1) + (n-1) \cdot friends(n-2)$

$n=4$ a, b, c, d 10

a, b, c, d

a, b, cd

a, c, bd

a, d, bc

b, c, ad

b, d, ac

c, d, ab

ab, cd

ac, bd

ad, bc

a b c d

a is single b c d

a can be paired up (n-1) me se koi ek

n-2 log bache hai unka ans

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$$f(n) = f(n-1) + (n-1) * f(n-2)$$

n=1	n=2	n=3	n=4
a	a, b	c, a, b	d, c, a, b
	ab	c, ab	d, c, ab
		ca, b	d, ca, b
		cb, a	d, cb, a
			da, b, c
			da, bc
			db, a, c
			db, ac
			dc, a, b
			dc, ab

	0	1	2	3	4	5
dp	X	1	2	4	10	26

$$dp[i] = dp[i-1] + (i-1) * dp[i-2]$$

$$if(i <= 2)\ dp[i] = i$$

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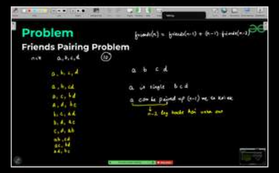

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Problem

Count Derangements

$$n!\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}\cdots\pm\frac{1}{n!}\right)$$

Party → n log aaye sab umbrella leke aaye

dance vance kiya,

saare 'n' log koi doosra umbrella le gaye → no. of ways.

3 postcards a, b, c

c / A	a / B	b / C
b / A	c / B	c / C

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